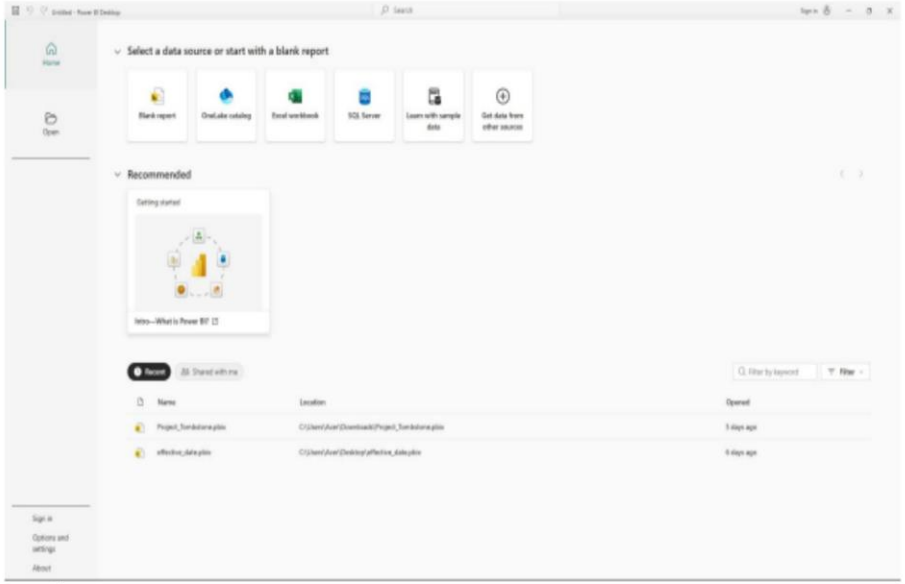
IS 107 · BUSINESS INTELLIGENCE

Introduction to Business Intelligence & Power BI

Harry M. Dajunos

Instructor: Joseph A. Vistal

PART 1 · STEP 1 — OPEN POWER BI DESKTOP



Part 1 | Step 2: Load the Dataset

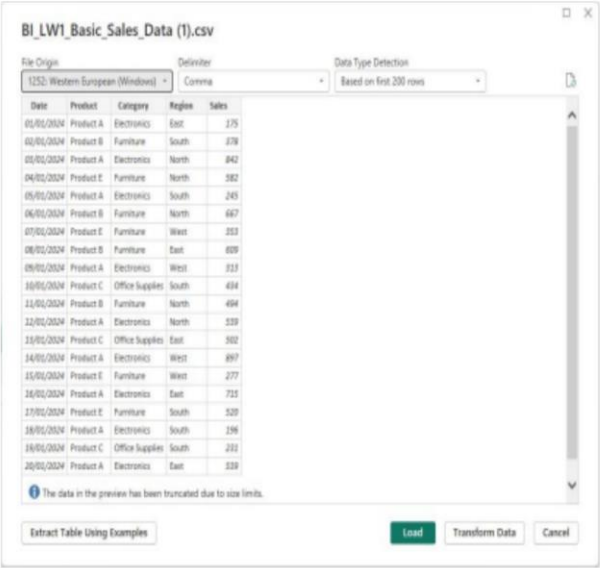
What to do

1. Launch Power BI Desktop.
2. On the splash screen, choose **Excel workbook** or **Get Data from other sources**.

3. Navigate to ***BI_LW1_Basic_Sales_Data.csv*** and click Load.

Part 1 | Step 2: Load the Dataset — Power BI Desktop Home Screen

PART 1 · STEP 3 — VERIFY DATA IN DATA VIEW



Part 1| Step 3 Verify Data in Data View

Are all columns visible?

Yes ✓

"Date" formatted as Date?

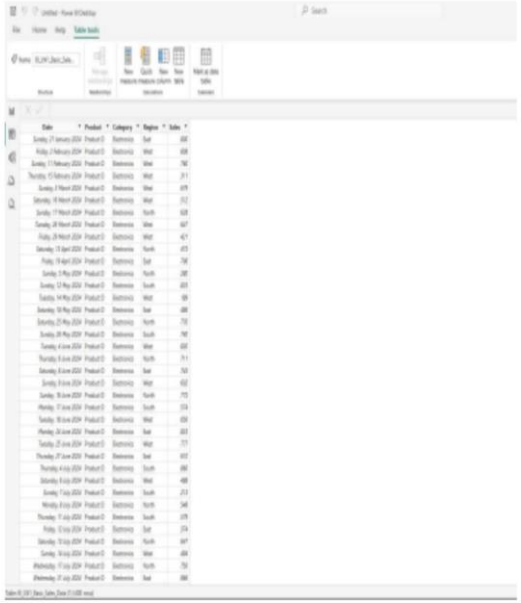
Yes ✓

"Sales" formatted as Decimal Number?

No — it is Whole Number

Part 1 | Step 3: CSV Preview Dialog — BI_LW1_Basic_Sales_Data.csv

PART 1 · TABLE VIEW — ALL 1,000 ROWS LOADED



Creating Auto-Generated Visuals

Part 3 | Step 1 Quick Visualization

Are all columns visible?

Answer: Yes

Is “Date” formatted as Date?

Answer: Yes

Is “Sales” formatted as Decimal Number?

Answer: No

1,000

Total Rows

5

Columns

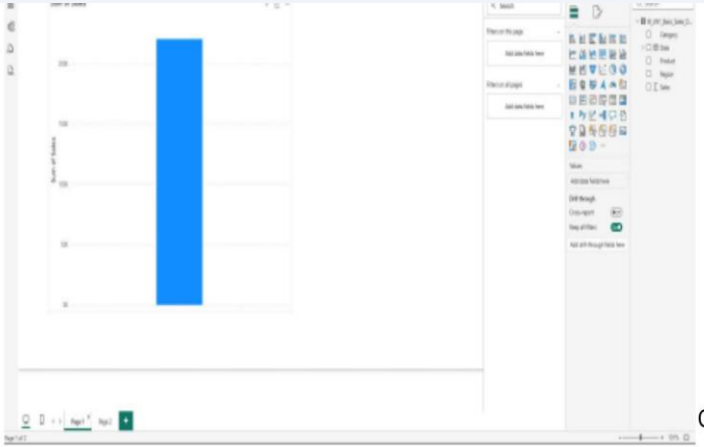
Date ✓

Date Type

Whole No.

Sales Type

Part 1 | Step 3: Table/Data View showing all 5 columns



What type of chart was created?

Answer: 100% Stacked Column Chart

What does it show?

Answer: Sum of Sales

Creating Auto-Generated Visuals

Part 3 | Step 2 Create a Sales by Region Chart

Findings

Chart Type Created:

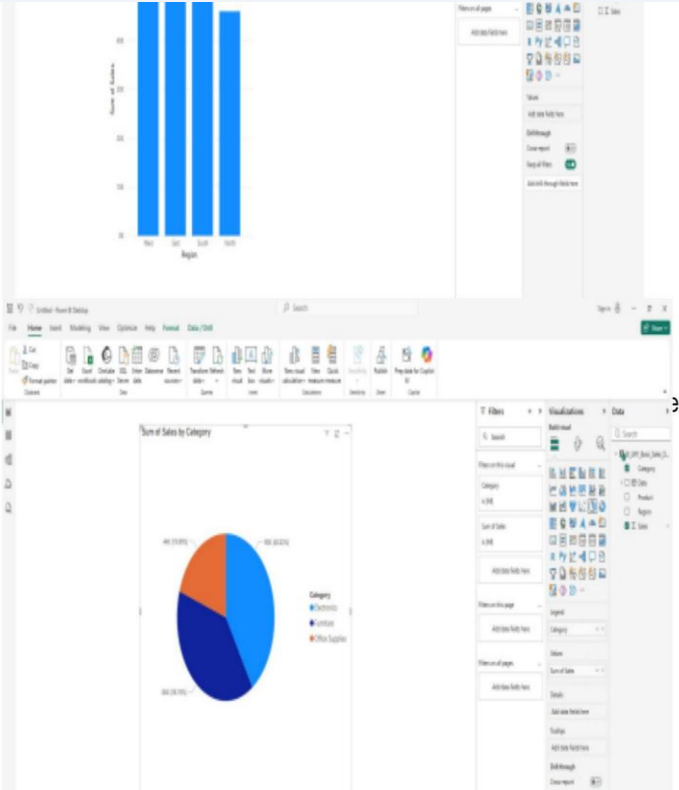
100% Stacked Column Chart

What does it show?

Sum of Sales

Part 3 | Step 1: Auto-generated 100% Stacked Column Chart

PART 3 · STEPS 2 & 3 — SALES BY REGION AND BY CATEGORY



Which region has highest sales?

Answer: West Region

eating Auto-Generated Visuals

Which category dominates?

Step 2 – By Region

Which region has highest sales?

→ **West Region**

Step 3 – By Category

Which category dominates?

→ **Electronics**

Step 3 – By Category

Is the distribution balanced?

→ **No**

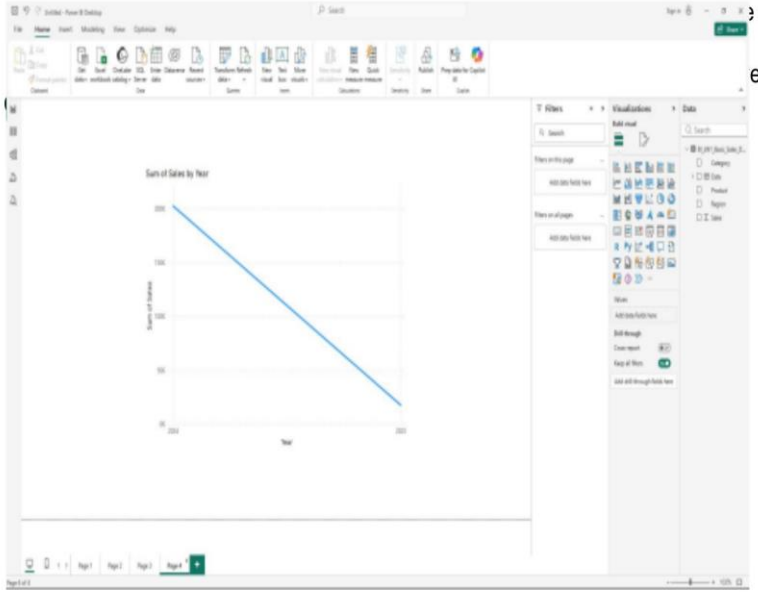
Part 3 | Steps 2–3: Bar chart (Regions) and Pie chart (Categories)

PART 3 · STEP 4 — SALES BY YEAR (LINE CHART)

Answer: Electronics

e distribution balanced?

er: No



Is there growth?

Answer: None

Any noticeable trend?

Answer: None

Line Chart Analysis

Is there growth?

→ **None**

Any noticeable trend?

→ **None — flat/declining**

Sales pattern

→ **Decreasing over time**

Part 3 | Step 4: Sum of Sales by Year — declining trend visible

BASIC DATA INSIGHT INTERPRETATION

Part 2 · Summary Findings

Top Revenue Region

West

Highest total sales across all regions

Best Product Category

Electronics

Dominates the sales pie chart (>40%)

Sales Consistency

Not Consistent

Varies significantly across dates

Business Recommendation

Invest in North

Expand operations in North region

Q1. What are the five columns in the dataset?

→ Date, Product, Category, Region, Sales

Q2. What data type is assigned to "Sales"?

→ Whole Number

Q3. Which Power BI view shows raw data?

→ Table View

Q4. What chart type best shows trends over time?

→ Line Chart

Q5. What aggregation is automatically applied?

→ SUM

Q1. Which region has the highest total sales?

→ West

Q2. Which category has the lowest performance?

→ Office Supplies

Q3. Are sales increasing, decreasing, or stable?

→ Decreasing

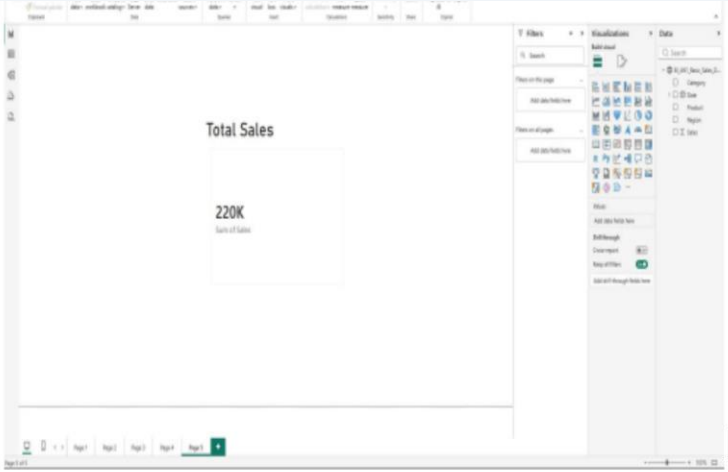
Q4. If you were a manager, which region to prioritize?

→ North

Q5. Provide one actionable recommendation.

→ Launch a targeted Office Supplies marketing campaign in the North region.

ENHANCEMENT · TASK 1 — CARD VISUALIZATION



What is the total sales amount?

Answer:220k

ENHANCEMENT SECTION

Advanced Exploration

Task 2 Add Slicer

Card Visual Result

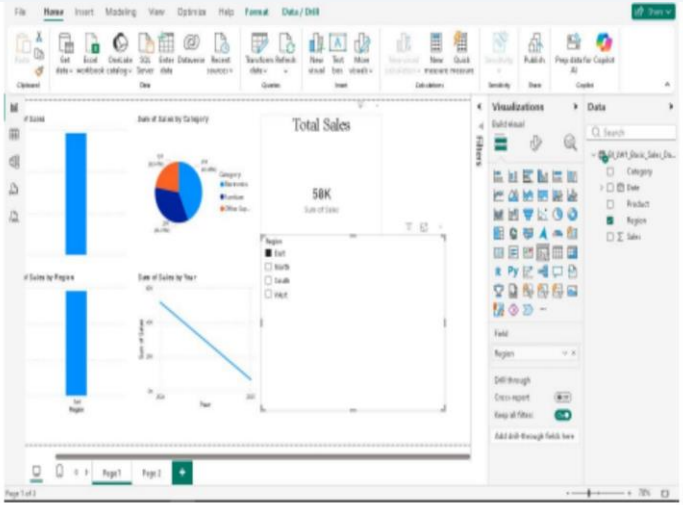
Total Sales Amount:

220K

Sum of Sales

A Card visual gives an instant single-number KPI summary.

Enhancement | Task 1: Card Visualization showing Total Sales = 220K



What happens to other visuals when you click a region?

Answer: The other visual will change.

Why is filtering important in BI?

Answer: Filtering is important in BI because it transforms overwhelming data into actionable insights.

ENHANCEMENT

SECTION

Advanced Exploration

Task 3 Sort Sales

What happens when you click a region?

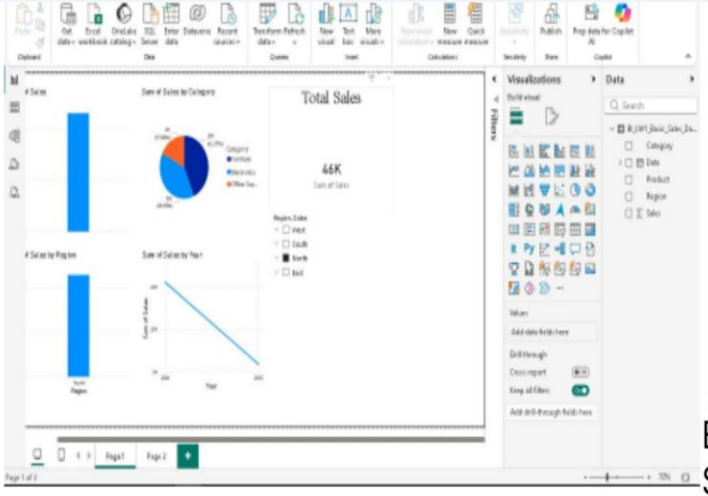
All other visuals update and filter to show only that region's data.

Why is filtering important in BI?

Filtering transforms overwhelming data into targeted, actionable insights.

Enhancement | Task 2: Slicer by Region — East selected, 58K total shown

ENHANCEMENT · TASK 3 — SORT SALES



Does sorting improve readability?

Answer: No

Why?

Answer: The sorting only rearrange the region in the card in descending order but the value still the same.

ENHANCEMENT SECTION

Advanced Exploration

Task 4 Identify Outliers

Which region is significantly higher or lower?

Sort Analysis

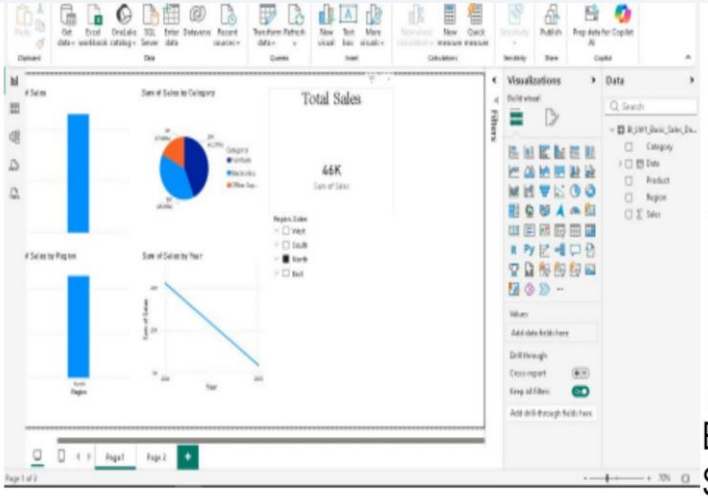
Does sorting improve readability?

No

Why?

Sorting only rearranges the regions in descending order, but the values themselves remain the same — no new insight is gained.

Enhancement | Task 3: Sorted by Sales descending — North slicer active, 46K shown



Does sorting improve readability?

Answer: No

Why?

Answer: The sorting only rearrange the region in the card in descending order but the value still the same.

ENHANCEMENT SECTION

Advanced Exploration

Task 4 Identify Outliers

Which region is significantly higher or lower?

Outlier Findings

Significantly HIGHER:

West Region

Significantly LOWER:

North Region

Explanation:

West region has significantly more buyers than North. This gap suggests an opportunity to grow North region's customer base.

Enhancement | Task 4: Outlier identification — West vs. North regional performance

KEY TAKEAWAYS

What We Learned

- Power BI Desktop allows quick loading and inspection of CSV/Excel datasets.
- Data type verification is critical — Sales was Whole Number, not Decimal.
- West Region leads in total revenue; Electronics dominates by category.
- Filtering with Slicers enables dynamic, targeted data exploration.
- Actionable: launch an Office Supplies campaign in the North region.